

Detecting Diabetes Misinformation: Leveraging Medical Expertise and NLP Models for Effective Detection

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Abstract—The proliferation of health-related misinformation on social media, particularly regarding diabetes, poses a substantial threat to public health decision-making and awareness. This study addresses the challenge of identifying false information about diabetes on Arabic social media platforms. A comprehensive dataset was collected directly from these platforms, with the verification of data authenticity conducted by medical professionals who accurately labeled the information. We subsequently evaluated various machine learning models for their effectiveness in detecting misinformation, achieving notable accuracy. Furthermore, the study incorporated an advanced Large Language Model (LLM), specifically GPT-4, to assess its proficiency in identifying fake news relative to the evaluations by our medical team. The findings highlight the efficacy of integrating human expertise with artificial intelligence to enhance the detection and mitigation of misleading health information.

Index Terms—Diabetes, Arabic, Medical Staff, Machine Learning, LLM.

I. INTRODUCTION

Diabetes, a chronic condition, is characterized by insufficient insulin production or the body's inability to effectively utilize insulin. Type 1 Diabetes is an autoimmune disorder necessitating lifelong insulin therapy, while Type 2 Diabetes typically develops due to genetic and lifestyle factors, resulting in insulin resistance or reduced secretion. Gestational Diabetes occurs during pregnancy. Globally, Over millions of adults aged 20-79 have diabetes, with projections indicating that it will rise in the future [1]. Low and middle-income countries bear the majority of this burden, with millions of deaths annually attributed to diabetes. Economic costs have surged to USD billions, with millions adults at high risk of Type 2 Diabetes due to Impaired Glucose Tolerance (IGT). Diabetes

prevalence is escalating, notably in South Asia and the Middle East, with Arab nations facing alarming rates, Children are also affected. Misinformation on social media exacerbates the issue, prompting the need to have a Diabetes Fake News Detection System (DFNDS) to empower individuals with accurate health information.

In recent years, the proliferation of fake news in healthcare has emerged as a significant challenge with potentially life-threatening implications, prompting researchers and practitioners to explore various strategies for its management. In [2] the authors stress the urgent need to mitigate misinformation's spread in healthcare, This study emphasizes the critical implications of fake news in healthcare and proposes practical solutions like dedicated websites and recommended systems to overcome this problem. Using evolutionary algorithms, it achieves moderate accuracy in identifying fake news, calling for further research into managing healthcare information. The authors of [3] explore the role of emotional awareness in enhancing the detection of fake health news, presenting innovative methodologies to improve accuracy in identifying misinformation in the healthcare domain. The authors of [4], their research introduce a fake news detection system that utilizes ontology to debunk myths, particularly in the medical domain. By focusing on enhancing Natural Language Processing (NLP) systems' interpretability through Description Logic (DL), the study aims to improve the identification of inconsistencies in texts, thereby advancing the field of fake news detection in healthcare. The authors of [5] examine the interplay between fake news detection, health literacy, and vaccine acceptance. It advocates for public education interventions to bolster information literacy and mitigate the impact of misinformation on public health. Introducing a method for analyzing news dissemination, also the authors

in [6] offer a promising avenue for the automated detection of fake news. Through heuristic analysis, the study addresses challenges posed by misinformation in healthcare. The authors [7] provide insights into machine-learning techniques for fake news detection, particularly in the context of COVID-19. It identifies ongoing efforts to combat misinformation and outlines challenges in detection methodologies. Finally, the authors of [8] show us the ability of healthcare professionals and students to discern fake news. The study underscores the influence of individual perceptions and emphasizes the importance of information literacy and critical thinking in fake news detection. Our research addresses a gap in the detection of misinformation about diabetes on non-English social media platforms, specifically focusing on Arabic discussions. We have developed a robust dataset, meticulously labeled by experienced medical professionals, that serves as a foundation for testing various machine learning models. Moreover, we compare these traditional models against advanced Large Language Models (LLMs) like GPT4 to assess their effectiveness in identifying and classifying false claims about diabetes.

The rest of the paper is organized as follows: In Section II, we discuss our Dataset collection and Preparation. Section III describes the employed system model. The findings from our empirical analysis across all datasets are presented and discussed in Section IV. Finally, Section V concludes the paper with a summary of key insights and conclusions drawn from the research.

II. DATASET COLLECTION AND PREPARATION

In this section, we will detail our proposed methodology Figure 1, which encompasses the processes of data collection, data cleaning, and labeling by healthcare professionals.

A. Data Collection

We've implemented robust data collection strategies focusing on gathering diverse and accurate sources of information related to diabetes. Leveraging web scraping techniques, we've accessed a broad spectrum of content encompassing news articles, forum discussions, and user posts from social media platforms like Facebook and Twitter. Our observations reveal that a significant portion of the information on Twitter pertains to Saudi Arabia, with contributions from other countries such as Egypt, Sudan, Jordan, and others. On the other hand, Facebook provides substantial data from countries like Egypt and the United Arab Emirates. Using Selenium in our Python script, we efficiently extract relevant data from these platforms, primarily in the form of text content or attributes, which we then organize and store in Excel sheets. This meticulous approach, combining Selenium's capabilities with Pandas' data manipulation functionalities, ensures the seamless extraction and organization of information from various web sources, offering valuable insights into prevalent narratives and trends surrounding diabetes.

B. Data Cleaning

In the data cleaning process, we systematically review each row of the scraped social media data from the Excel

file. We begin by removing any mentions of usernames or handles, as well as hyperlinks, as they often don't contribute to the analysis. Additionally, we filter out posts that primarily focus on marketing links or personal interactions unrelated to diabetes. This ensures that the dataset contains only relevant information about diabetes. By meticulously curating the content and eliminating noise, such as non-diabetes-related posts, marketing materials, and personal conversations, we streamline the dataset to be ready for labeling.

C. Healthcare consultant and labeling

In our quest to enhance our understanding of diabetes within the Arabic nation, we extend beyond basic data collection by tapping into the expertise of both local and international professionals. We actively collaborate with Egyptian physicians from various regions, including Cairo, Fayoum and Beni Suef, as well as those working overseas, particularly in Italy. This collaboration enriches our understanding of global best practices and current trends in diabetes care. Additionally, we consult with renowned nutrition experts in Egypt to thoroughly investigate the dietary aspects of diabetes prevention and management. Our detailed approach also includes the careful organization of data, as demonstrated by our dataset, which contains over 10,000 carefully selected entries related to diabetes. Each piece of data is evaluated by three medical staff experts and labeled as true (T) or false (F) using a majority voting strategy to ensure the highest accuracy for the label, who also ensure its accuracy against established medical standards and evidence-based practices. Figure 2 shows two samples from the dataset while figure 3 shows the distribution of the T and F samples from the medical expert process. This meticulous process guarantees the quality and dependability of our dataset, supporting informed research and decision-making in diabetes healthcare and studies.

III. SYSTEM MODEL

This section explores our proposed system model and the various models we implemented to classify social media posts accurately. We detail the methodologies applied across various model architectures, which are presented in Figure 4.

A. Long Short Term Memory (LSTM)

The First model in our research employs a Long Short Term Memory LSTM neural network architecture for the task, leveraging Python libraries such as TensorFlow and scikit learn for implementation. Initially, we loaded our dataset, where each entry consists of textual content related to diabetes along with corresponding labels indicating the authenticity of the information. The preprocessing phase involves removing non-linguistic characters, tokenization, and encoding the labels into numerical values using Label Encoder from Scikit learn. Following preprocessing, the textual data is tokenized using the Tokenizer class from TensorFlow, which converts text into sequences of integers based on word frequency. To ensure uniform input lengths, padding is applied to sequences using the pad sequence's function. The LSTM model architecture

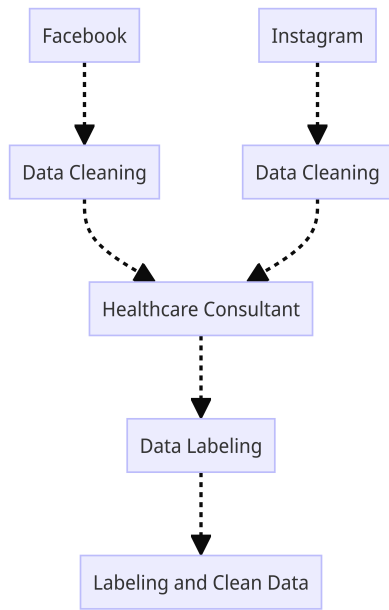


Figure 1. Data Collection Methodology

Posts	Label	Source
فحص العيون سنويا للكشف عن أي ضرر قبل أن يؤثر على حدة البصر	T	Twitter
وجود الزلازل في البول لمرضى السكري هو أحد علامات أن وظائف الكلى بدأت تتأثر.	T	Twitter

Figure 2. Sample of Dataset

is then defined using TensorFlow's Sequential Application Programmable Interface API comprising an embedding layer, a spatial dropout layer for regularization, an LSTM layer with dropout, and a dense layer with sigmoid activation for binary classification, figure 5 shows the detailed architecture of the model. During model training, early stopping is employed to prevent overfitting, monitor the validation loss and terminate training if no improvement is observed for a specified number of epochs. After training, the model's performance is evaluated

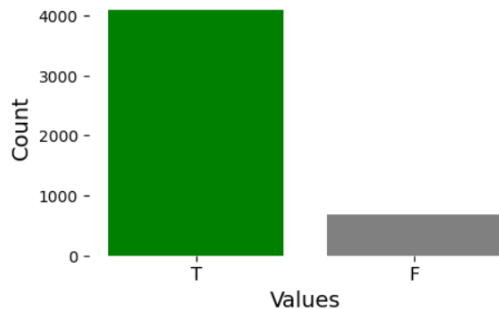


Figure 3. Distribution of T and F in the Dataset

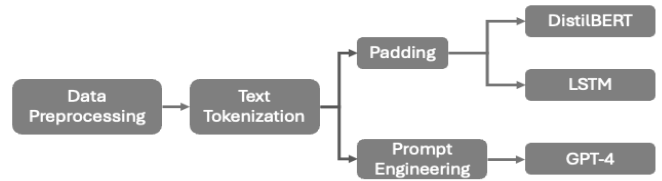


Figure 4. proposed system model

on the test set, yielding an accuracy of approximately 89.87%. Additionally, classification metrics such as precision, recall, and F1-score are computed to assess the model's performance comprehensively. Furthermore, a confusion matrix is generated to visualize the model's performance in distinguishing between true and false news instances.

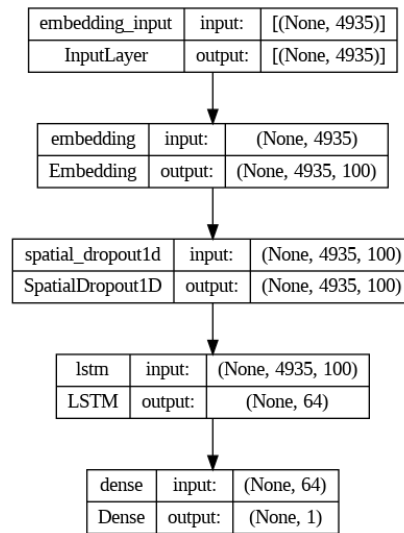


Figure 5. LSTM Model Structure

B. BERT (Bidirectional Encoder Representations from Transformers)

The second model in our study utilizes the BERT Bidirectional Encoder Representations from Transformers [9] architecture, specifically the Distil BERT variant, with dependencies from the Transformers library for BERT models. The model is trained and evaluated on our dataset, where each entry consists of textual content along with corresponding labels denoting the authenticity of the information. The pre-processing phase involves converting the textual labels into integer representations True to 1, False to 0 and splitting the dataset into training and testing sets. Tokenization is performed using the Distil BERT tokenizer, which encodes the text into input IDs and attention masks, ensuring uniform input lengths of 512 tokens. The training loop encompasses multiple epochs, during which the model is trained on the training dataset batches. Adam W optimizer is employed for parameter optimization, with early stopping implemented to prevent over fitting. Checkpoints of the model are saved periodically to

track the best performing model based on validation loss. Upon completion of training, the model is evaluated on the test dataset, and predictions are made using the trained model. These predictions are then utilized to construct a confusion matrix, providing insights into the model's classification performance. The confusion matrix visually represents the distribution of true positive, false positive, true negative, and false negative predictions.

C. GPT₄

In The Third model, We incorporated a model leveraging AZURE Open AI with GPT4. Facilitating the evaluation of the authenticity of textual information pertaining to diabetes management. Within the framework of this model, we employed the AZURE Open AI library to interact with GPT4, a state-of-the-art language generation model developed by Open AI. The model's implementation involved loading our dataset and iterating through each row to prompt GPT4 with specific queries regarding the accuracy of the provided information. Notably, the model's methodology did not encompass extensive preprocessing steps, focusing primarily on the interaction with GPT4 to obtain binary responses indicative of the authenticity of the queried content.

IV. RESULT

In this section, we present the outcomes of the various models proposed in our study.

A. Long Short-Term Memory (LSTM)

The LSTM model achieved an accuracy of approximately 89.87% on the test dataset, indicating a powerful performance in binary classification between true T and false F labels. examining the classification report and confusion matrix, it is evident that the model's precision for the false class (F) is lower compared to the true class (T). Specifically, the precision for class F is around 90%, indicating that among the instances predicted as false, approximately 90% are false. However, the recall for class F is low at 34%, suggesting that the model struggles to correctly identify a sizable portion of false instances. This imbalance in precision and recall for class F could be attributed to the class imbalance in the dataset, as there are significantly fewer instances of false labels compared to true labels. Moreover, analyzing the top 20 predictions from the test set reveals that the model tends to predict most instances as true (T), with high probabilities for true instances and comparatively lower probabilities for false instances. However, it occasionally misclassified false instances as true, as indicated by the presence of false instances with high predicted probabilities. This behavior aligns with the observations from the classification report and confusion matrix, highlighting the model's tendency to favor the majority class.

B. Result Bert

The Distil BERT model trained for sequence classification exhibits robust performance in accurately classifying Arabic

text data into binary categories of 'True' and 'False'. Throughout the training process, the model demonstrates a consistent decrease in both training and validation losses across multiple epochs, indicating effective learning and generalization capabilities. Despite terminating early due to the early stopping criterion, the model achieves impressive accuracy on the test dataset, reaching approximately 96%. The classification report further confirms the model's effectiveness, with high precision, recall, and F1-score values for both the 'True' and 'False' classes. Specifically, the model achieves a precision of 0.92 for the 'False' class and 0.97 for the 'True' class, indicating its ability to make accurate predictions while minimizing false positives and false negatives. The balanced performance across all metrics suggests that the model can effectively discriminate between true and false instances within the dataset. the confusion matrix provides additional insights into the model's performance, showing a minimal number of miss classification. Out of 1097 instances in the test set, the model miss classifie only 21 instances as 'False' when they are actually 'True', and vice versa, highlighting its robustness and reliability in distinguishing between the two classes. And the model successfully identifies the label of a randomly selected text from the test dataset, showcasing its ability to generalize well to unseen data. In the given example, the model accurately predicts the label of the text, demonstrating its proficiency in understanding and classifying Arabic text related to medical topics Figure 6 presents a detailed comparison between the Distil BERT and the LSTM model.

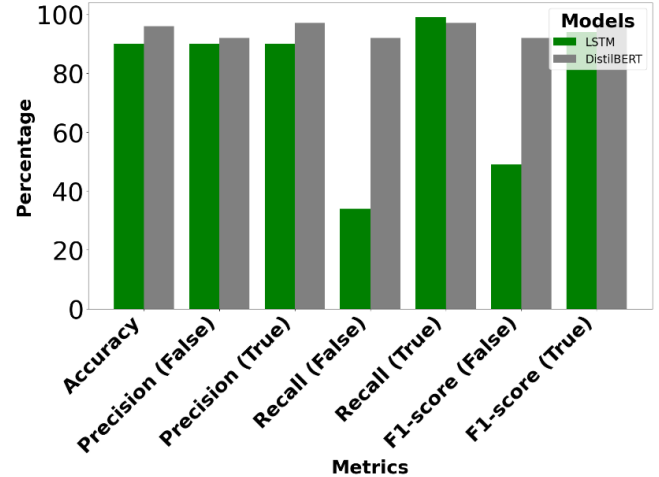


Figure 6. Comparison of LSTM and DistilBERT Models

C. Result Gpt-4

Upon comparing the actual labels of the dataset with the predictions generated by GPT4 model using AZURE Open AI a minor difference was observed. Specifically, when considering the 'healthcare consultant' class, there was a slight variance between the predicted labels and the true labels in the dataset figure 7. Additionally, the model's responses demonstrate its awareness of its limitations in verifying certain

types of content. Responses such as 'Sorry, I can't help verify the content and Sorry, I can't provide an answer without specific information indicate the model's ability to acknowledge situations where it lacks context or data to provide a reliable response. This demonstrate the model's self awareness and its cautious approach in providing accurate information.

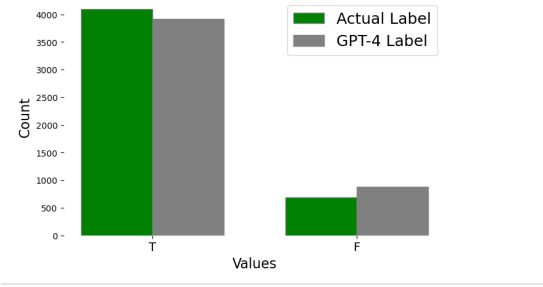


Figure 7. Result of GPT

V. DISCUSSION AND CONCLUSION

The detection of fake news related to diabetes is a critical issue that demands attention to aid the community in identifying false information on social media. In this study, we compiled a comprehensive dataset from various social media platforms concerning diabetes, which was meticulously verified by experienced medical staff to ensure accurate labeling. To enhance the effectiveness and reliability of our models, we propose a framework for continuous data collection and model training, involving medical experts in the loop. This framework allows for regular updates and verification of user-submitted data, ensuring that our models stay current with emerging misinformation trends and evolving narratives. Using this verified dataset, we developed three distinct models to classify fake news related to diabetes from social media posts or tweets. These models assist in determining the reliability of the information. Among the models tested, the DistilBERT model demonstrated superior precision and recall, outperforming the LSTM model. Meanwhile, the GPT-4 model achieved results comparable to those of medical expert consultations. However, it is important to note that the dataset might contain inherent biases based on the sources and regions from which the data was collected. These biases can affect the model's performance and its ability to generalize across different populations and types of misinformation. To overcome this issue, we are targeting specific regions and languages in our current framework. To generalize the model effectively, we will need to collect a significantly larger dataset from diverse regions and languages

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